

**Pharmacology: Sedative-Hypnotics
(Summary of Drugs and Practice Questions)**

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Sedative-Hypnotic Drugs

Barbiturates: These are GABA independent drugs. They bind to the chloride channel and facilitate GABA action (duration of channel opening). However, they can directly promote chloride conductance. Not limited by GABA, only dose.

- Pentobarbital, Secobarbital, Phenobarbital, Amobarbital Thiopental
- No “ceiling effect”
- Can induce full respiratory arrest (fatal)
- Provide supportive care and alkalinize urine
- Use: anxiety and insomnia (fallen out of use), seizures (phenobarbital), induction of anesthesia (thiopental).
- Barbiturates are known inducers of CYP450
- Lipid soluble drugs (thiopental) are redistributed from brain.
- Elderly and those with liver disease may have an increase in half-life.
- Highly sedative with a hangover effect
- Tolerance and withdrawal = physical dependence
- Taper slowly. Highly abused (pentobarbital mostly because it’s short acting)

Benzodiazepines (BDZ): Potentiates GABA by increasing frequency of chloride channel opening. Needs GABA to open channel.

- Antidote is Flumazenil (benzo antagonist)
 - “Ceiling effect” Safer than barbiturates.
- No respiratory depression, unless combined with other CNS depressants.

Use: anxiety, sedative-hypnotic, insomnia, anterograde amnesia, muscle relaxant, anticonvulsant.

Chlordiazepoxide and Diazepam (long-acting drugs): Used for alcohol detoxification.

Alprazolam (short acting): panic disorder. Most abused as it is short acting

Lorazepam, Clonazepam, Diazepam: Seizure disorder.

Midazolam: anesthesia; Remimazolam (newest; rapid onset and offset)

Estazolam and Flurazepam: insomnia

Chlordiazepoxide, diazepam, flurazepam (metabolites make drug long acting).

Oxazepam, temazepam, and lorazepam are the safest for the elderly. Bypass phase I metabolism (Only undergo conjugation).

Ethanol: Potentiates GABA (can induce full respiratory depression, especially if combined with other drugs such as barbiturates, benzodiazepines, and opioids)

- Euphoria (at small doses)

- Reward center (dopamine)
- Dilated cardiomyopathy (chronic use). Also, hepatotoxicity, disrupts immunity, endocrine
- Zero-order elimination kinetics
- Fetal Alcohol Syndrome (teratogen)—thin lips, microcephaly, cognitive effects
- Wernicke's (thiamine deficiency); give thiamine followed by glucose.

Ethanol Intoxication

- Slurred speech, nystagmus, stupor, or coma.
- Hypotension and tachycardia
- Hypoglycemia, lactic acidosis, hypokalemia, hypomagnesemia
- Treatment: supportive care, ventilation
- 80mg/dL BAC (intoxication)

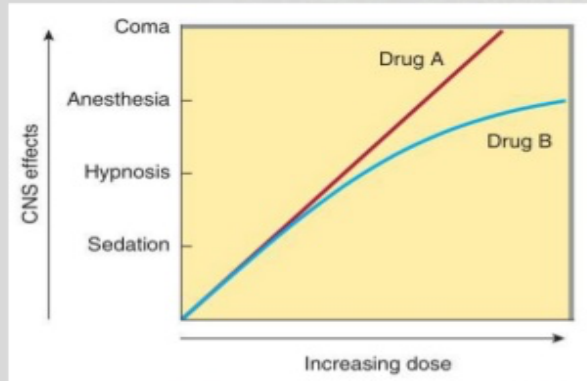
Ethanol Overdose, Withdrawal, Abstinence

- Overdose: maintain airway, provide supportive care
- Symptoms of withdrawal: CNS stimulation, seizures “delirium tremens” BENZO must be administered. Long-acting Benzo (ex. Chlordiazepoxide).
- Abstinence: Naltrexone (opioid blocker); Acamprosate (restores GABA/glutamate balance lost by chronic ethanol consumption)

Other drugs:

- Clonidine: α_2 agonist. Decreases the sympathetic overdrive associated with withdrawal. Sedation, decreased BP, and heart rate are adverse effects of clonidine.
- Disulfiram: inhibits aldehyde dehydrogenase; deterrent to ethanol abuse. Metallic taste is an adverse effect.
- Fomepizole: alcohol dehydrogenase inhibition (manages antifreeze poisoning)

Dose-response curves for two hypothetical sedative-hypnotics



- Drug A:
 - An increase in dose higher than that needed for hypnosis may lead to a state of general anesthesia.
 - With higher doses, the drug will depress the respiratory and vasomotor centers which leads to coma.
 - Drug A is an example of alcohol and barbiturates.
- Drug B:
 - needs greater doses to achieve CNS depression.
 - Drug B is an example of benzodiazepines and newer hypnotics.

Flumazenil: Antagonist at benzodiazepine (BDZ) binding site at the GABA receptor

- Blocks actions of BDZ and “zz” drugs, but not for other sedative- hypnotics When given IV, it increases recovery following BDZ administration in anesthetic procedures.
- Adverse effects include agitation, confusion, and withdrawal symptoms in those who are chronic BDZ users.

Drugs for insomnia, narcolepsy, and anxiety

Drug	MOA	Use	Adverse Effect
Zolpidem Eszopiclone Zaleplon “Non-Benzodiazepines” “zz” drugs	GABA potentiation Benzodiazepine-like action on Cl⁻ channels	Minimal effect on REM sleep Shorter duration of action Less hangover effect and daytime sedation Don’t take unless you can devote 7-8 hours to sleep. Target specific GABA_A subunit.	Anterograde amnesia Hallucinations Delusions Impaired judgement Sleepwalking Depressant action potentiated by other CNS depressants and alcohol
Ramelteon Tasimelteon	MT-1, MT-2 melatonin receptor agonist Melatonin observed to regulate circadian rhythm	Insomnia Not addictive (not a controlled substance) Possible antidepressant effect	
Suvorexant Lemborexant Daridorexant	Orexin antagonist Orexin is a NT in lateral hypothalamus	Insomnia Other uses under investigation	Daytime sedation
Amphetamine Dextroamphetamine Methylphenidate	CNS stimulation, inhibits reuptake of NE and DA, increases release of these catecholamines	Narcolepsy ADHD	Sympathomimetic effects (cardio stimulation, insomnia, weight loss)

	from nerve terminals		
Modafinil Armodafinil	greater DA vs NE reuptake inhibition and release Increased orexin Reduced GABA Increased histamine (enhanced wakefulness and reduced appetite)	Narcolepsy	Less anxiety and anorexia than amphetamines Skin rash
Sodium oxybate	Sodium salt of gamma-hydroxybutyrate (GHB), metabolite of GABA. Agonist of GABA_B receptor	Narcolepsy with cataplexy.	N/V, confusion, not to be administered with ethanol or sedative hypnotics (increased risk of respiratory depression). REMS (FDA) Short half-life May be combined with daytime stimulant.
Propranolol	Beta1, beta2 antagonist	Blocks SANS contribution to anxiety “Situational anxiety”	CNS depression, fatigue, bradycardia Asthma, hypoglycemia, and

			PVD because of B₂ blockade
Buspirone	5-HT_{1A} agonist (partial)	Generalized anxiety	No sedation or addiction liability Headache, dizziness, nervousness Several weeks for full effect. Paired with Benzodiazepines short-term (until buspirone effect)

****Note: OTC sleep aids (Sominex[®], Unisom[®] Tylenol-PM[®],) contain diphenhydramine as an active ingredient. Recall the MOA and adverse effects associated with first-generation H₁ blockers (they are lipophilic enough to cross the blood brain barrier) and muscarinic antagonist adverse effects (dryness, constipation, mydriasis, urinary retention, increase AV and SA node conduction). Drugs with muscarinic blockade are not indicated for those with benign prostatic hypertrophy (BPH), glaucoma, and the elderly (increase delusional effects). Trazodone (antidepressant) is a serotonin blocker and reuptake inhibitor which can induce sleep as it has H₁ antagonist potential.

Second generation antihistamines: loratadine, fexofenadine, and cetirizine are **not** sedative because they do not cross the blood brain barrier. They also **lack antimuscarinic effects**.

Questions

1. Which drug potentiates GABA, with minimal hangover effect if taken as directed?
 - A. Alprazolam
 - B. Flurazepam
 - C. Zolpidem
 - D. Ramelteon
 - E. Phenobarbital

2. A 50-year-old female presented with complaints of difficulty sleeping but did not want a drug with addiction potential. The physician decided on a medication which activates melatonin receptors. Which does this describe?
 - A. Diazepam
 - B. Diphenhydramine
 - C. Phenobarbital
 - D. Ramelteon
 - E. Zaleplon

3. A 27-year-old female was started on medication for generalized anxiety. Since this drug took at least four weeks for full effect, the physician decided to add another medication for that short period of time to prevent any additional acute effects. Which of the following does this describe?
 - A. Alprazolam
 - B. Buspirone
 - C. Propranolol
 - D. Modafinil
 - E. Zolpidem

4. A 17-year-old female violinist was to perform at a concert this evening. She was concerned that her nervousness would interfere with her ability to play. Which of the following is effective for her currently?
 - A. Buspirone
 - B. Clonazepam
 - C. Propranolol
 - D. Methylphenidate
 - E. Zolpidem

5. Which of the following describes the mechanism of action of lorazepam?
- A. Activation of the 5HT₁ receptor
 - B. Blockade of dopamine reuptake
 - C. Increased frequency of chloride channel opening
 - D. Increased duration of chloride channel opening
 - E. Inhibition of glutaminergic N-methyl-D-aspartate receptors
6. A 56-year-old male alcoholic entered a treatment facility. Which of the following would reduce his intense cravings for a drink, by blockade of the mu (opioid) receptor?
- A. Acamprosate
 - B. Diazepam
 - C. Sodium oxybate
 - D. Naltrexone
 - E. Thiamine
7. An 80-year-old female complained to her physician of difficulty falling asleep. Although the use of a benzodiazepine was considered, the physician knew that one could cause the most “hang-over” effects and the possibility of falls in this patient, since it was long-acting. Which of the following does this describe?
- A. Alprazolam
 - B. Diazepam
 - C. Lorazepam
 - D. Oxazepam
 - E. Zolpidem
8. A 20-year-old male presented to his physician with extreme anxiety, not managed by psychotherapy. The physician decided on a medication that although not habit-forming, would take at least several weeks to have a full effect. Which of the following does this describe?
- A. Alprazolam
 - B. Buspirone
 - C. Lorazepam
 - D. Propranolol
 - E. Zaleplon

9. A 26-year-old female with a history of depression, was brought to the emergency room extremely sedated, after having been found on the floor with an empty bottle of clonazepam next to her. Which of the following should be administered?

- A. Acamprosate
- B. Diazepam
- C. Flumazenil
- D. Naltrexone
- E. Thiamine

10. A 50-year-old female was to have a surgical procedure performed. The anesthesiologist was to begin sedation with thiopental. Which of the following describes the mechanism of this drug?

- A. Agonist at the MT-1 receptor
- B. Increased duration of chloride channel opening
- C. Increased frequency of chloride channel opening
- D. Inhibition of the H₁ receptor
- E. Inhibition of norepinephrine and serotonin reuptake

11. A 50-year-old male presented with complaints of insomnia and when he does sleep, he wakes up many times during the night. He decided to begin an over-the-counter medication known to increase sedation, without addiction liability. Which of the following does this describe?

- A. Agonist at the beta-1 receptor
- B. Increased duration of chloride channel opening
- C. Increased frequency of chloride channel opening
- D. Blockade of the H₁ receptor
- E. Inhibition of norepinephrine and serotonin reuptake

Review Question:

Which drug for the management of MS binds CD20?

- A. Fingolimod
- B. Interferon
- C. Natalizumab
- D. Oxybutynin
- E. Rituximab

Answers

- 1. C**
- 2. D**
- 3. A**
- 4. C**
- 5. C**
- 6. D**
- 7. B**
- 8. B**
- 9. C**
- 10. B**
- 11. D**

Review Question

E. Rituximab